Cybersecurity

Historical ciphers

Cybersecurity

Attacking Transposition Ciphers

- Simple transposition cipher alter dependencies between consecutive characters, but preserves the frequency distribution of each letter
- Anagramming
  - If 1-gram frequencies match English frequencies, but other *n*-gram frequencies do not, probably transposition
  - Rearrange letters to form *n*-grams (1st digram then trigram) with highest frequencies

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Attacking Transposition Ciphers

- ***Example:***
  - Plaintext is "`HELLOWORLD`"
  - Encryption key is e=(1 3 5 7 4 2 9 6 8 10)
  - Ciphertext is `HLOOLELWRD`
- Frequencies of 2-grams beginning with H
  - HE 0.0305
  - HO 0.0043
  - HL, HW, HR, HD < 0.0010
- Frequencies of 2-grams ending in H
  - WH 0.0026
  - EH, LH, OH, RH, DH ≤ 0.0002
- Implies E follows H

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Attacking Transposition Ciphers

- Factor – to identify possible rectangles.
  - 64 letters could be 8 x 8 or 16 x 4 or four 4 x 4.
- Arrange – into rectangle.
- Count the vowels – 40% vowels, 12% e's
  - Rows and columns should have ~ 40% vowels.
- Rearrange – rows and/or columns.
  - Digrams, trigrams, probable sequences guide the rearrangements.

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Attacking Transposition Ciphers

```
Message: ETTUH OMEAW EXETE STDEH TIGYT HLKHS MAODT O
Letters = 36  Possible squares 6x6, 9x4, 12x3

E M E E T M   3          E W E K   2
T E T H H A   2          T E H H   1
T A E T L O   3          T X T S   0
U W S I K D   2          U E I M   3
H E T G H T   1          H T G A   1
O X D Y S O   2          O E Y O   4
                         M S T D   0
3 3 2 2 0 3              E T H D   1
                         A D L O   2

                         5 3 3 3
```

- Analysis:
  - In the 6x6 square, the 40% rule gives 2-3 vowels per row or column.
  - The rows seem better, so rearrange the columns.

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Attacking Transposition Ciphers

```
E M E E T M    E M E E T M    M E E T M E    M E E T M E    M E E T M E
T E T H H A    T E T H H A    E T H H A T    E T T H A H    A T T H E H
T A E T L O    E A T T L O    A T T L O E    A T E L O T    O T E L A T
U W S I K D    S W U I K D    W U I K D S    W U S K D I    D U S K W I
H E T G H T    T E H G H T    E H G H T T    E H T H T G    T H T H E G
O X D Y S O    D X O Y S O    X O Y S O D    X O D S O Y    O O D S X Y
```

- Analysis
  - AE (row 3) don't go, swap cols 1 & 3.
  - MEET ME in row 1 by moving col 1 to end
  - HH (row 2) don't go, swap cols 3 & 6
  - THA becomes THE (row 2) by swapping cols 1 and 5.

`Message: MEET ME AT THE HOTEL AT DUSK WITH THE GOODS`

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Substitution Ciphers

- Simple substitution cipher changes letters in plaintext to produce ciphertext
- Simple substitution cipher is called also ***mono-alphabetic substitution***

Cybersecurity

Attacking simple substitution ciphers

- Statistical analysis
  - Simple substitution cipher alter the frequency of the individual plaintext characters, but …
  - doesn't alter the frequency distribution of the overall character set
  - Thus, letter frequency analysis helps breaking the cipher

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How to break the substitution cipher?

- Use statistical patterns of the language.
- For example: the frequency tables.
- Texts of 50 characters can usually be broken this way.

Letter	Frequency
E	0.127
T	0.097
I	0.075
A	0.073
O	0.068
N	0.067
S	0.067
R	0.064
H	0.049
C	0.045
L	0.040
D	0.031
P	0.030
Y	0.027
U	0.024
M	0.024
F	0.021
B	0.017
G	0.016
W	0.013
V	0.008
K	0.008
X	0.005
Q	0.002
Z	0.001
J	0.001

Figure 7 - Frequency Table

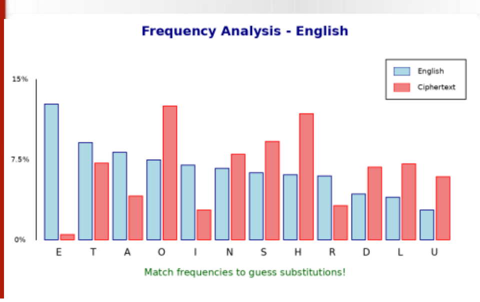
Cybersecurity

Frequency of digrams in English text (%)

di-gram	frequency	di-gram	frequency
AN	1.81	ON	1.83
AT	1.51	OR	1.28
ED	1.32	RE	1.90
EN	1.53	ST	1.22
ER	2.31	TE	1.30
ES	1.36	TI	1.28
IN	2.30		

Cybersecurity

Histogram of letter frequencies



Cybersecurity

Substitution Ciphers

- **Having a large key space is necessary to prevent an exhaustive key search, but it is not sufficient to guarantee the security of a cipher system.**
- In other words (Martin Hellman's, to be precise…)
- **A large key is not a guarantee of security**
  but a
  small key is a guarantee of insecurity.

Attacking simple substitution ciphers

- Conclusion
  - Simple substitution ciphers are easy to recognize and analyze
  - How to improve simple substitution ciphers?
  - In other words, how to defeat letter frequency analysis?
    - Polygram ciphers: Playfair cipher
    - Polyalphabetic substitution ciphers: Vigenere cipher

Polygram substitution ciphers

- Simple substitution cipher substitutes one character by other character
- Polygram substitution cipher substitutes groups of characters by other groups of characters
- Examples
  - Sequences of 2 plaintext characters (digrams) may be replaced by other digrams
  - Sequences of 3 plaintext characters (trigrams) may be replaced by other trigrams
- Playfair cipher is an example of polygram substitution ciphers

Playfair cipher: Example

Playfair Cipher

Keyword: MONARCHY

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

Rules:

1. Same row → replace with letter to the right
2. Same column → replace with letter below
3. Rectangle → replace with letter in same row, other corner

Example: GL → EQ FH **→ PF**

Attacking Playfair cipher

- Playfiar cipher alters the frequency of the individual plaintext characters and alters the frequency distribution of the overall character set because each letter may be replaced by other.
- However, digram frequency analysis helps breaking the cipher

Polyalphabetic substitution cipher

- ***Polyalphabetic substitution cipher*** is a block cipher with block length t over an alphabet A:
  - The key space consists of all ordered sets of t permutation $(p_1, .., p_t)$, where each p_i is defined on the set A
  - Encryption the message $m=(m_1,\ldots,m_t)$ under the key $e=(p_1,\ldots,p_t)$
  - The decryption key associated with e
- Example: Vigenère cipher

Vigenère cipher

- A={A,…,Z}
- $t=3$
- $e=(p_1,p_2,p_3)$ encryption key
  - p_1: maps each letter to the letter 3 positions to its right in A
  - P_2: maps each letter to the letter 7 positions to its right in A
  - P_3: maps each letter to the letter 10 positions to its right in A
  - This means that e= CHK
- m = THI SCI PHE RIS CER RAI NLY NOT SEC URE
- *c* = WOS VJS SOO UPC FLB WHS QSI QVD VLM XYO

Cybersecurity

Vigenère's Autokey Cipher

- Uses mixed alphabets and the text itself as the key.

```
Encryption:

Therearesomethirty (MSG=col)
Xtherearesomethirt (KEY=row)
OZGUUAOUVIZNWZMYDP (Crypto)

Decryption:

OZGUUAOUVIZNWZMYDP(MSG=ltr)
XthereareS..     (KEY=row)

O in row X gives column T
...
V in row E gives column S
```

```
   ABCDEFGHIJKLMNOPQRSTUVWXYZ
A | ZEBRAFISHCDGJKLMNOPQTUVWXY
B | EBRAFISHCDGJKLMNOPQTUVWXYZ
C | BRAFISHCDGJKLMNOPQTUVWXYZE
D | RAFISHCDGJKLMNOPQTUVWXYZEB
E | AFISHCDGJKLMNOPQTUVWXYZEBR
F | FISHCDGJKLMNOPQTUVWXYZEBRA
G | ISHCDGJKLMNOPQTUVWXYZEBRAF
H | SHCDGJKLMNOPQTUVWXYZEBRAFI
I | HCDGJKLMNOPQTUVWXYZEBRAFIS
J | CDGJKLMNOPQTUVWXYZEBRAFISH
K | DGJKLMNOPQTUVWXYZEBRAFISHC
L | GJKLMNOPQTUVWXYZEBRAFISHCD
M | JKLMNOPQTUVWXYZEBRAFISHCDG
N | KLMNOPQTUVWXYZEBRAFISHCDGK
O | LMNOPQTUVWXYZEBRAFISHCDGKL
P | MNOPQTUVWXYZEBRAFISHCDGKLM
Q | NOPQTUVWXYZEBRAFISHCDGKLMN
R | OPQTUVWXYZEBRAFISHCDGKLMNO
S | PQTUVWXYZEBRAFISHCDGKLMNOP
T | QTUVWXYZEBRAFISHCDGKLMNOPQ
U | TUVWXYZEBRAFISHCDGKLMNOPQT
V | UVWXYZEBRAFISHCDGKLMNOPQTU
W | VWXYZEBRAFISHCDGKLMNOPQTUV
X | WXYZEBRAFISHCDGKLMNOPQTUVW
Y | XYZEBRAFISHCDGKLMNOPQTUVWX
Z | YZEBRAFISHCDGKLMNOPQTUVWXY
```

Cybersecurity

Attacking Vigenère cipher

- Doesn't preserve symbol frequencies
  - In the example: E is encrypted to O and L
- However, it's not significantly more difficult to cryptanalyze
- The approach:
  - determine the period t (*i.e.*, key length)
  - Ciphertext can be divided into t groups (group i consists of those ciphertext letters derived using permutation p_i)
  - Letter frequency analysis to be done on each group

Cybersecurity

Attacking Vigenère cipher - Kasiski

- Great idea:
  - Certain sequence of letters appears many times in a text
  - A ciphertext that uses Vigenere will have the sequence of letters (encrypted) showing in many places
- So:
  - Record distances between repetitive sequences of letters greater than 3 letters
  - Good candidate for the number of alphabets (key size) is the gcd of the recorded distances

Plaintext: TOBEORNOTTOBE
Key: NOWNOWNOWNOWN
Ciphertext: GCXRCNACPGCXR

Cybersecurity

Attacking Vigenère cipher - Friedman's IC

Index of Coincidence

- Index of coincidence – the number of times two identical letters occur in the same position in two adjacent texts.
- William F. Friedman (Father of American cryptography).

$$IC = \sum_{i=1}^{c} \frac{n_i(n_i-1)}{N(N-1)/c}$$

Language	Normalized IC (1/26)
Plain English	0.067
German	0.079
Caesar English	0.065
Simple English Substitution	0.065
Uniform distribution	0.0385

Malay	0.085286
Dutch	0.079805
Japanese	0.077236
Hebrew	0.076844
German	0.076667
Spanish	0.076613
Arabic	0.075889
French	0.074604
Portuguese	0.074528
Finnish	0.073796
Italian	0.073294
Danish	0.070731
Norwegian	0.069428
Greek	0.069165
English	0.066895
Swedish	0.064489
Serbo-Croatian	0.064363
Russian	0.056074
Random	0.038461

Cybersecurity

Attacking Vigenère cipher - Friedman's IC

- Gives the probability that 2 randomly selected letters are the same
- For plain English, prob. 2 letter are same:
  - $p_0^2 + p_1^2 + \ldots + p_{25}^2 \approx 0.065$, where p_i is probability of i^{th} letter
- Then for simple substitution, $I \approx 0.065$
- For random letters, each $p_i = 1/26$
  - Then $p_0^2 + p_1^2 + \ldots + p_{25}^2 \approx 0.03846$
- Then $I \approx 0.03846$ for poly-alphabetic substitution with a very long keyword

Cybersecurity

Attacking Vigenère cipher - Friedman's IC

- How to use this to estimate length of keyword in Vigenere cipher?
- Suppose keyword is length k, message is length n
  - Ciphertext in matrix with k columns, n/k rows
- Select 2 letters from same columns
  - Like selecting from simple substitution
- Select 2 letters from different columns
  - Like selecting random letters

Cybersecurity

Attacking Vigenère cipher - Friedman's IC

Plaintext		Ciphertext
UNTILMODERNTI		FRVBFDSOITG
MESCRYPTOGRAP		NZQPWEKSGX
HYREFERREDALM		ZKTTJYCCIHXL
OSTEXCLUSIVELYT		IIOENFIJXPBEE
OENCRYPTIONWH	Alg. Vigenere	OJMGINRNFIY
ICHISTHEPROCESS	Password: LECTURE	GTRJKMZRYAC
OFCONVERTINGO		TLTWVAYGVZ
RDINARYINFORM		GGLMFJNSPOY
ATIONINTOUNINT		IXTRIHLUMYET
ELLIGIBLEGIBBERI		RCEJZVOTNZS
SHDECRYPTIONIST		YMPMILRTRVX
HEREVERSEINOTH		FCMRMDEYXM
ERWORDSMOVIN		MFGKCJLOIEK
GF		SGXTSPBMKLP
		VGOYIWPMPH
		NYICAQKXJQZ
		ZKGAW

Cybersecurity

Attacking Vigenère cipher - Friedman's IC

Ciphertext

FRVBFDSOITG
NZQPWEKSGX
ZKTTJYCCIHXL
IIOENFIJXPBEE
OJMGINRNFIY
GTRJKMZRYAC
TLTWVAYGVZ
GGLMFJNSPOY
IXTRIHLUMYET
RCEJZVOTNZS
YMPMILRTRVX
FCMRMDEYXM
MFGKCJLOIEK
SGXTSPBMKLP
VGOYIWPMPH
NYICAQKXJQZ
ZKGAW

Ciphertext

L	E	C	T	U	R	E
F	R	V	B	F	D	S
O	I	T	G	N	Z	Q
P	W	E	K	S	G	X
Z	K	T	T	J	Y	C
.	.	.	.	.	.	.

Monoalphabetic cipher

Cybersecurity

Attacking Vigenère cipher - Friedman's IC

Index of Coincidence (IC)

IC = Probability that two random letters are the same

English text	Random text
IC ≈ 0.065	**IC ≈ 0.038**

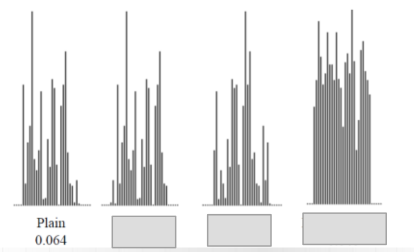
Usage for Vigenère Cipher:

- IC ≈ 0.065 → Monoalphabetic (key length = 1)
- IC ≈ 0.038 → Polyalphabetic with long key
- IC between → Polyalphabetic, use to find key length
- Test different period values, highest IC = key length

Cybersecurity

Attacking Vigenère cipher - Friedman's IC

IC – Example



Cybersecurity

Attacking Vigenère cipher

The Chi-square Test

- Used for comparing and observed frequency distribution with an expected distribution.
- Goodness-of-fit statistic – the smaller the better.

$$X^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i},$$

Cybersecurity

Attacking Vigenère cipher

Automated Algorithm

- Assumes: long enough polyalphabetic cipher text
- For each possible keyword length, 2, 3, 4, ..., *k*
  - Divide the text into *k* columns.
  - Compute the average the IC for the columns.
  - Select the IC that is closest to 0.067
- For each of the *k* columns
  - For each possible shift, 1..26.
  - Compute the Chi-square value.
  - Select the minimum as the correct Caesar alphabet.

Cybersecurity

Homophonic Coding

- A modification of the Simple Substitution Cipher

- The idea is to replace some plaintext letters with different ciphertext characters in order to confuse the ciphertext character frequency statistics.

Cybersecurity

Homophonic Coding

- A possible homophonic code is as follows:
- Use a character alphabet of 1000.
- Use our table of letter frequencies:
  - use 82 different characters to encode A
  - use 15 different characters to encode B
  - use 1 character to encode J, Q, Y and Z
  - etc.
- How well do you think such a code will hold out against single letter frequency analysis?

- Do you think that such a code is suitable for adoption in a practical environment?

Cybersecurity

Product ciphers

- Simple transposition and substitution don't provide a very high level of security
- ***product cipher*** is a combination of many transformations (transposition and substitution)
- ***product ciphers*** are a way to obtain strong ciphers
- We'll call a ***round*** a composition of a substitution and a transposition.

Cybersecurity

Historical Ciphers